INCH-POUND
MIL-W-24753(SH)
22 JANUARY 1991

MILITARY SPECIFICATION

WASHER, HIGH PRESSURE, DIESEL POWERED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a portable, high pressure (2,800 to 3,000 pounds per square inch (lb/in^2), hot and cold, fresh water washer system, diesel powered.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-F-320	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shape

MILITARY

MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-S-12651	Sprayers, Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC/NA FSC 4940

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MIL-P-15024	Plates, Tags and Bands for Identification of Equipment
MIL-P-15024/5	Plates, Identification
MIL-L-19140	Lumber and Plywood, Fire Retardant Treated
MIL-C-22750	Coating, Epoxy-Polyamide
MIL-P-23377	Primer Coatings, Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-C-83286	Coating, Urethane, Aliphatic Isocyanate, for Aerospace Applications

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR

Code of Federal Regulations Title 29 Part 1910	Occupational Safety and Health Standards

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(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 **Non-Government publications.** The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN WELDING SOCIETY (AWS)

D1.1	Structural Welding Code	Steel; (DOD adopted)
D1.2	Structural Welding Code	Aluminum; (DOD adopted)

(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 **General.** The washer system shall consist of a diesel engine; pump; water heater; freshwater and chemical storage and dispensing system; wand assembly; wet/dry vacuum; generator; operational and safety controls; and trailer mounting for use with aircraft and cargo handling equipment commonly known as yellowgear.

3.3 **Standard commercial product.** The washer shall, as a minimum, be in accordance with the requirements of this specification and shall be the contractor's standard commercial product. Additional or better features which are not specifically prohibited by this specification, but which are a part of the contractor's standard commercial product, shall be included in the washer being furnished.

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3.4 **Materials.** Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials shall be of the same quality used for the intended purpose in commercial practice. Parts which are subject to corrosion in a marine environment shall be either aluminum flame sprayed, zinc galvanized, or constructed of stainless steel, brass, bronze, or other corrosion resistant material.

3.4.1 **Recovered materials.** Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4.2 **Asbestos and cadmium.** The use of asbestos and cadmium/cadmium plating is prohibited.

3.5 **Construction.** The unit shall permit easy accessibility for maintenance and service in the fleet. The construction shall be such as to prevent conditions hazardous to personnel or deleterious to equipment.

3.5.1 **Dimension and weight.** The washer system shall not exceed a length of 11 feet, a width of 5 feet, and a height of 5 feet. The washer system shall not exceed a dry weight of 2,800 pounds and an aggregate weight of 5,000 pounds.

3.5.2 **Power unit.** Power shall be supplied by a diesel engine that operates with JP-5 fuel in accordance with MIL-T-5624 and has horsepower to provide continuous service with a variable load. The engine shall be complete with starter, generator, voltage regulator, remote controls, 12-volt battery with waterproof battery box and cables, and high temperature, low oil level and low oil pressure automatic cutoff controls. A tachometer, hourmeter, or combination tachourmeter shall be furnished. When specified (see 6.2), the battery furnished with the unit shall be dry charged, without electrolyte, with sealed caps to prevent the intrusion of atmospheric moisture. Ventilation shall be provided to assure adequate engine cooling and ventilation of exhaust gases.

3.5.3 **Tanks.** Separate fresh water, chemical and fuel tanks shall be provided and shall have a capacity sized for a minimum of 25 minutes of washer operation. A means of heating the fresh water to a temperature of 195 degrees Fahrenheit (degrees F) within 5 minutes of washer start-up, and maintaining the water temperature between 175 and 195 degrees F during washer operation shall be provided. Fluid lines leading from the tanks shall include a filter system consisting of replaceable or cleanable type filters. Tanks shall be provided with a gravity drainage system.

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3.5.4 **Pump.** The pump shall be driven by a diesel engine and shall deliver fresh water at the specified volume and pressure. The pump shall be furnished with a pressure relief valve, factory set to 1-1/2 times the rated pressure, on the discharge side of the pump. A gravity drainage system shall be provided.

3.5.5 **Pressure relief dump valve.** A pressure relief dump valve shall be located downstream of the pump and upstream of the water heater. The relief pressure shall not exceed 110 percent of rated discharge. The over pressured discharge shall dump into the water supply tank.

3.5.6 **Wand.** The spray wand shall consist of a gun, with trigger and internal parts, high pressure lance and chemical dispensing tube.

3.5.6.1 **Gun.** A pistol grip gun shall be equipped with a trigger to control discharge to the lance. There shall be a trigger guard and an automatic safety back-out, in the off position, to prevent accidental discharge of high pressure water.

3.5.6.2 **Lance.** The lance shall be 3 feet ± 2 inches long, fabricated from schedule 80 stainless steel. An insulated support sleeve or handle shall protect operating personnel from burns. The high pressure lance shall be equipped with nozzles of three different spray patterns; 15, 25, and 45 degrees. The nozzles shall be corrosion resistant, attached to the high pressure lance by a positive locking method and provide steady or pulsating jets or both.

3.5.7 **Hose.** The hose shall be constructed of an extruded rubber tube, multiple layers of wire braid and a cover compounded for maximum abrasion resistance. The hose shall be 1 inch inside diameter (ID); be able to pass water/detergent solutions at 200 degrees F, 3,000 lb/in^2 (12,000 lb\in^2 burst); and have reusable brass or corrosion resistant steel fittings. Each hose section is to be 50 feet in length with a minimum of two sections provided and the unit shall have a manual take up reel or other means of storage. The hose is to be non-collapsing.

3.5.8 **Wet/dry vacuum.** The washer shall be equipped with a wet/dry vacuum that runs off the 120-volt alternating current (Vac) double outlet provided with the washer unit. The vacuum shall have a minimum 12-gallon recoverable capacity; quick coupling snap on hose connection; auto shut-off to prevent overflow, and a minimum lift of 40 inch water rating (see 4.7.7).

3.5.9 **Chemical dispensing system.** The chemical dispensing system shall insert chemicals into the fresh water flow downstream of the pump to avoid contamination of the pump. Chemical valve controls shall be provided on both the unit and wand assembly.

3.5.10 **Trailer.** The trailer shall be twin axel, rigid frame with tires, brake and steerable front axel capable of being moved over 4 inches high protrusions encountered on decks. The tongue of the trailer shall be equipped with a 4-inch lunette eye that will mate with a universal pintle hitch of a tow tractor. It shall provide an enclosed weathertight environment for all component

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...e made to ensure that water does not collect on any surfaces. Lunettes for ... provided on four corners. A static grounding chain shall be provided from the main frame to the deck.

3.5.10.1 Tires. The tires shall be solid or foam filled to 45 lb/in^2 pneumatic equivalent, load range C or D.

3.5.10.2 Control panel. Gauges, switches, meters, and controls shall be mounted on a single control panel located inside the enclosure. Permanently marked gauges indicating water temperature, water pressure, washer operation elapsed time, oil level, fuel level, water level and chemical level shall be included.

3.5.10.3 Doors. The enclosure shall include sliding doors that provide easy access to the control panel. The doors shall be such that they may be secured in either the open or closed position.

3.6 Electrical.

3.6.1 Switch. The diesel engine shall have a waterproof, keyed ignition starter switch located on the diesel engine control panel and a waterproof abort switch(s) for stopping of the engine on each remote control panel provided. Switches shall operate easily and reliably and be clearly marked on the control panel(s).

3.6.2 Grounding. Electrical equipment shall be grounded to avoid shock hazards.

3.6.3 Generator. If a diesel engine driven generator is installed for powering electrically powered accessories, it shall provide 120 Vac, 60 hertz, maximum 2,000 watts of electrical power to a 15 ampere ground fault interrupter (GFI) overload protected circuit.

3.7 Safety. Mechanical or electrical parts that are of such nature or so located as to become a hazard to operating or maintenance personnel shall be enclosed or properly guarded.

3.7.1 Safety controls. The cleaner shall be equipped with all mechanical and electrical controls and devices necessary for safe and efficient operation. Such controls shall include the following:

a. Safety control to shut down the water heating accessory in the event the temperature of the heating accessory exceeds a predetermined safe limit due to partial or total interruption of the water flow to the coil. This control may be responsive to coil temperature, coil expansion, water flow rate, water storage levels, or any equally reliable indicator.

b. Mechanical or electrical pressure control device to prevent the pressure in the heating coil and delivery hose from exceeding a predetermined safe limit.

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c. Automatic fuel shut-down device in event of ignition interruption

d. A safety-relief valve for relieving water.

3.8 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.9 Cleaning, treatment, and painting. Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be painted in accordance with the following procedure: Exterior surfaces shall be corrosion treated by first applying MIL-P-23377, class 1 Epoxy polyamide primer; then painted with a finish coat of MIL-C-22750 class 1 paint that is yellow in color. The yellow is FED-STD-595, yellow No. 13538. The final coat is a clear coat of MIL-C-83286 urethane sealer. Sufficient times or heat processes shall be allowed between coats to ensure the previous coat is thoroughly dry before adding the next coat.

3.10 Identification plate. Each unit shall bear a manufacturer's name plate in accordance with MIL-P-15024 and MIL-P-15024/5 for normal service. The plate shall contain the manufacturer's name, model number, contract number and serial number of the unit.

3.10.1 Instruction plate. When any special or important procedure is to be followed in operating and servicing the equipment, an instruction plate describing such procedure shall be affixed to the unit. Plates shall be of copper or zinc base-alloy, CR aluminum, adequately secured. Lettering shall be black on a light background and sized and arranged as specified in MIL-STD-1472.

3.11 Performance.

3.11.1 Capacity. The washer shall produce a flow of not more than 5 gallons per minute, and a nozzle discharge pressure of 2,800 to 3,000 lb/in^2 (see 4.7.1).

3.11.1.1 Continuous operation. The washer shall operate continuously as specified (see 3.11.1), for not less than 48 hours (see 4.7.2).

3.11.2 Safety. Mechanical or electrical parts that are of such nature or so located as to become a hazard to operating or maintenance personnel shall be enclosed or properly guarded. The cleaner shall conform to the Code of Federal Regulations, Title 29, Chapter XVII, Part 1910 (see 4.7.3).

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3.11.3 **Shock.** The washer wand and delivery hose assembly shall operate satisfactorily (see 3.11.1 and 3.11.1.1) with no breakage, fracture, and leakage after a fall of 6 feet onto a smooth concrete floor (see 4.7.4).

3.11.4 **Electromagnetic interference control.** The washer shall meet the electromagnetic interference control requirements and test limits specified in MIL-STD-461 and MIL-STD-462 (see 4.7.5).

3.11.5 **Noise.** The washer, while operating, shall not exceed a sound level of 85 decibels A-weighted (dBA) (see 4.7.6).

3.11.6 **Brake.** The trailer brake system shall immobilize the trailer on a 15-degree incline (normal direction, non-skid deck) with supply tanks full as a tow motor is normally used to move the washer a brake is not necessary. Wheels can be chocked if unit is unattended in conjunction with chain falls to deck. Otherwise the brake requirement will be a draw bar pull of 50 percent of the aggregate weight.

3.12 **Workmanship.**

3.12.1 **Steel fabrication.** The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which are deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than specified by the design. Manufacturing processes shall be done neatly and accurately. Bends shall be made by controlled means to ensure uniformity of size and shape.

3.12.2 **Bolted connections.** Bolt holes shall be accurately punched or drilled and the burrs shall be removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.12.3 **Riveted connections.** Rivet holes shall be accurately punched or drilled and the burrs shall be removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the diameter of the rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.12.4 **Welding.** Welding procedures shall be in accordance with AWS D1.1 and D1.2 for structural welding of steel and aluminum. The surface of parts to be welded shall be free from corrosion, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.3). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as a part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirement, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article inspection shall be performed on one complete washer. This inspection shall include the examination of (see 4.6) and tests (see 4.7). The first article may be either a first production item or a standard production item from the contractor's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.4 Quality conformance inspection. Quality conformance inspection shall include the examination (see 4.6) and tests (see 4.7) and table I for all machines offered for delivery at one time.

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TABLE I. *Classification of defects.*

Item	Defect	Major	Minor
Materials	Components or parts not fabricated/used as specified (see 3.4 and 3.4.1)	X	
	Asbestos/cadmium materials used (see 3.4.2)	X	
Design/general	Washer does not contain all items specified (see 3.2 and 3.3)	X	
	Dimension and weight limits exceeded (see 3.5.1)	X	
	Power unit and accessories not as specified (see 3.5.2)	X	
	Tanks and their function not as specified (see 3.5.3)	X	
	Trailer not as specified (see 3.5.10)	X	
	Unit not properly prepared and painted (see 3.9)	X	
	Identification marking not as specified (see 3.10)	X	
	Instruction plates not as specified (see 3.10.1)	X	
	Workmanship is not as specified (see 3.12, 3.12.1, 3.12.2, 3.12.3, and 3.12.4)		X
Details of units or parts	Pump unable to perform as specified (see 3.5.4, 3.11.1, 3.11.1.1, 4.7, 4.7.1, and 4.7.2)	X	
	Pressure relief/dump value not as specified)see 3.5.5)	X	
	Spray wand not as specified (see 3.5.6, 3.5.6.1, 3.5.6.2, and 4.7.4)	X	
	Hose not as specified (see 5.5.7, 4.7.4, and 4.7.8)	X	
	Wet/dry vacuum not as specified (see 3.5.8 and 4.7.7)	X	
	Chemical dispensing, system not as specified (see 3.5.9, 4.7 and 4.7.1)	X	
	Tires not as specified (see 3.5.10.1)	X	
	Brakes not as specified (see 3.11.6)	X	
	Control gauges not as specified (see 3.5.10.2)	X	
	Doors not as specified (see 3.5.10.3)	X	
Electrical	Switches not as specified (see 3.6.1)	X	
	Electrical equipment not ground as specified (see 3.6.2, 3.5.10, and 3.7)	X	
	Generator not as specified (see 3.6.3 and 3.11.2)	X	
Safety	Safety controls not as specified (see 3.7, 3.7.1, 3.11.2, and 4.7.3)	X	
	Noise levels are exceeded (see 3.11.5 and 4.7.6)	X	
	Ground fault interrupter not as specified (see 3.6.3)	X	
	Electromagnetic interference control not as specified (see 3.11.4 and 4.7.5)	X	
	Asbestos and/or cadmium used (see 3.4.2)	X	

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4.5 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified herein.

4.6 Visual and dimensional. The washer shall be examined for conformance to the requirements specified herein. Any redesign or modification of the contractor's standard product to conform to specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall include all visual examinations and dimensional measurements. Nonconformance to any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.7 Operational tests.

4.7.1 Performance. The washer shall be connected and operated in accordance with the contractor's operating instructions. With the trigger depressed, the washer shall be operated at the rated discharge of not more than 5 gallons per minute at 2,800 to 3,000 lb/in^2 for 25 minutes. During this 25 minute period, the engine and pump shall be operated continuously, except that after each 5 minutes of operation, the trigger shall be released and the operation of the pressure reduction system observed for 30 seconds. Pressure shall be reduced to that of the inlet pressure of the pump. Upon completion of this test, no component of the washer (engine, pump, hose connections, control mechanism, and so forth) shall show evidence of failure, damage, or leakage.

4.7.2 Continuous operation. This test shall be performed at an ambient temperature of 70 $\pm$ 5 °F. With the trigger depressed the washer shall be operated at rated discharge of not more than 5 gallons per minute at 2,800 to 3,000 lb/in^2 for a total of 48 hours. After each 20 minutes of operation, the trigger shall be released and operation of the pressure reduction system observed for 30 seconds. Pressure shall be reduced to that of the inlet pressure of the pump. The water and fuel tanks shall be immediately refilled and the washer shall be restarted to continue the test. As a minimum, eight full cycles per day shall be performed until the machine has been operational for 48 hours. The water temperature, discharge pressure, and flow rate shall be recorded at the start and 20 minute period of each cycle. Upon completion of this test, there shall not be any failure of, damage to, or leakage by any component of the cleaner (engine, pump, hose connection, control mechanisms, and so forth).

4.7.3 Safety control. After the test of 4.7.2, the washer shall be tested to verify conformance (see 3.7). The conditions which activate the safety controls shall be simulated by interruption of water and fuel flows, as applicable, and for the pressure control test, by throttling the discharge.

4.7.4 Drop. The washer wand and delivery hose assembly shall be dropped two times from a height not less than 6 feet onto a smooth concrete floor. One drop shall impact on the handle and one drop shall impact on the wand. Upon completion of the two drops, the handle and wand shall be inspected for permanent deformation, breaks, and fractures. Either breakage or fracture of the handle or wand shall be cause for rejection. Permanent deformation of the handle and wand shall not be cause for rejection. The washer shall then be operated, and the handle and

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wand inspected for leaks. Leaks correctable by tightening of threaded connections shall not be cause for rejection. Leaks that cannot be corrected by tightening of threaded connections shall be cause for rejection. Failure of the cleaner to operate satisfactorily shall be cause for rejection.

4.7.5 **Electromagnetic interference control tests.** The washer shall be tested in accordance with MIL-STD-461, UMO3 for group III equipments.

4.7.6 **Noise.** Sound pressure measurements shall be taken at the normal position of the unit operator, stationed at the control panel, and shall be in accordance with 3.11.5.

4.7.7 **Vaccum test.** Vaccum pressure shall be measured by inserting a manometer into the vacuum line (see 3.5.8).

4.7.8 **Hose.** Each hose assembly shall be proof tested to 6000 psi and then inspected for signs of kinking, chafing, blistering, leaking, etc. The presence of any abnormality will constitute a failure.

4.8 **Inspection of packaging.** Sample packs, and the inspection of the preservation packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2 see 6.5.)

5.1 **Packaging requirements.** Diesel powered washers shall be preserved level A, C, or commercial, packed level A, B, or commercial as specified (see 6.2) and marked in accordance with MIL-S-12651 and shall include bar codes and applicable packaging acquisition options therein as specified (see 6.2). In addition, for Navy acquisitions, the following applies:

a. Navy shipboard stowage fire-retardant requirements:

(1) *Treated lumber and plywood.* Unless otherwise specified/When specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B	–	Type II - weather resistant Category I - general use
Level C	–	Type I – non-weather resistant Category I – general use

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(2) *Fiberboard.* Unless otherwise specified/When specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packaging forms shall meet the flamespread and the specific optic density requirements of PPP-F-320 amendments thereto.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The equipment covered by this specification is intended for use in heavy duty shipboard cleaning tasks such as arresting gear, catapult wells, cargo holds, well decks, bottoms of ships boats, topside cleaning, superstructure cleaning, weather decks, hangar decks and flight deck cleaning. This equipment should be operated by two men, one operating the pump, the other operating the wand.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
c. When first article inspection is required (see 3.1)
d. When a dry charged battery is required (see 3.5.2)
e. Test conditions, if other than as specified (see 4.5)
f. Level of preservation packing, marking required, and stowage fire-retardant requirements (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, Contract Data Requirements Lists (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.5.7	DI-E-2121	Certificate of Compliance	–

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6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When a first article inspection is required, the item should be a first article sample (see 3.1). The first article should consist of one complete washer (see 4.3). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Design criteria. All equipments, components and systems shall be in accordance with the applicable/appropriate UL, ASTM, ASME, SAE, ANSI, NEMA, etc. specifications and codes.

6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Subject term (key word) listing.

Lance
Wand
Wet/dry vacuum

Preparing activity:
Navy – SH
(Project 4940-N563)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-W-24753(SH)	2. DOCUMENT DATE (YYMMDD) 22 November 1991

3. DOCUMENT TITLE
WASHER, HIGH PRESSURE, DIESEL POWERED

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Poing of Contact (TPOC) Mr. D.C. Beeman (Code NAVSSES 074E) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: 443-7309	(2) AUTOVON 8-443-7309
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command SEA 5523 Department of the Navy Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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